

TB Resonator

DETAILED ENGLISH USER MANUAL

A playable eight-mode resonant body that turns incoming transients and sustained energy into tuned physical-model-like material.



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Host-visible endpoints documented: 7

1. Product purpose

A playable eight-mode resonant body that turns incoming transients and sustained energy into tuned physical-model-like material.

EQ can emphasize existing frequencies but cannot create the sustained behavior of a resonant object. Reverb creates space but not a defined musical body. TB Resonator excites eight stable modes so drums, noise, guitars, vocals, and effects can ring like strings, wood, metal, chambers, or tuned percussion.

What result it creates

- Tuned resonance from C1 to C7
- Continuous body movement from soft to string to metal
- Strike-to-drive excitation behavior
- Memory from short body to 30-second tail

Signal flow

Input excitation analysis -> Touch weighting -> eight modal resonators tuned by Tune and Body -> decay controlled by Memory -> Mix

2. Complete feature guide

Tune

Sets the resonant center from C1/32.7 Hz to C7/2093 Hz.

Body

Moves continuously through soft, string-like, and metallic modal relationships. It changes harmonicity and material impression.

Touch

Moves from strike-oriented excitation toward sustained drive. Low settings favor transients; high settings allow continuous energy to keep the body active.

Memory

Controls actual modal decay from 40 ms to 30 seconds. Long values can sustain across phrases and consume headroom.

Mix

Blends the resonant body with the source. At high Mix, the plug-in behaves more like an instrument excited by the input.

Programs

Programs cover chambers, strings, tuned drums, sympathetic guitar, vocal glass, bass wood, chimes, blooms, and driven drones.

3. Quick start

1. Set Tune to the song or desired object pitch.

2. Choose Body by listening to interval structure.
3. Set Touch for struck or sustained source behavior.
4. Set Memory for the required decay.
5. Raise Mix until the resonant body sits correctly.
6. Leave headroom for overlapping modes.

4. Practical workflows

Tuned drum

1. Use a low or mid Tune.
2. Set Touch toward Strike.
3. Use short-to-medium Memory.
4. Choose a less metallic Body and moderate Mix.

Expected result: The drum gains a clear, controlled musical body.

Ambient drone

1. Use a stable Tune.
2. Set Touch toward Drive.
3. Use long Memory and a metallic or string Body.
4. Automate Mix slowly.

Expected result: Sustained source energy excites a long evolving resonant instrument.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- Long Memory allows tails after transport stops.
- Resonant notes can conflict with the song key.
- High Mix and overlapping excitation can build peaks quickly.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.

- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 7 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Body VST3 ID 3029410	SOFT 0% to METAL 100%	STRING 50%	Automatable	Host-automatable control for Body. Its full range and default are shown in this table; use it with the related feature section above.
Bypass VST3 ID 773352680	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Memory VST3 ID 1069726977	40 ms to 30.0 s	1.10 s	Automatable	Sets how long the associated detector, envelope, reverb, or resonant process takes to recover or decay.
Mix VST3 ID 108124	0% to 100%	35%	Automatable	Sets the balance between the original signal and the complete processed path.
Program VST3 ID 1886548852	Init, Soft Chamber, Open String, Short Body, Tuned Drum, Sympathetic Guitar, Vocal Glass, Bass Wood, Harmonic Bloom, Metallic Bloom, Struck Chime, Driven Drone	Init	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Touch VST3 ID 110550847	STRIKE 0% to DRIVE 100%	NATURAL 50%	Automatable	Host-automatable control for Touch. Its full range and default are shown in this table; use it with the related feature section above.
Tune VST3 ID 3571704	C1 32.7 Hz to C7 2093 Hz	C3 130.8 Hz	Automatable	Sets the principal frequency, note, or spectral center used by this function.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.